MILITARY SPECIFICATION

BUSES, MOTOR: TRANSIT, AMBULANCE CONVERSION, FORWARD CONTROL, WITH HIGH HEADROOM, 30 AND 44 PASSENGER, 2 WHEEL DRIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers 30- and 44-passenger, forward driver control, two-wheel drive, diesel engine driven motor buses with ambulance conversion features. The bus body shall be a flat front type with high headroom. Vehicles shall have manufacturer's standard rear wheel drive, soft ride suspension, heavy duty bus chassis frame, and single front and rear axles. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The bus shall be of the type specified (see 6.2). Vehicle type, litter, and passenger capacity shall be as shown in table I.

* TABLE I. Types and passengers.

Type	Description
I	(See 6.5).
III	Ambulance conversion, 12 or 16 litter (see 3.5.13.4), 44 adult passenger
IV	Ambulance conversion, 12 litter, 30 adult passenger

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2310

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

L-S-300	- Sheeting and Tape, Reflective: Nonexposed Lens, Adhesive Backing.
A-A-393	- Extinguishers, Fire, Dry-Chemical (Portable).
W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting, and Starting.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).
LLL-F-1238	- Floor Covering, Linoleum.

STANDARDS
FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595	- Colors.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MIL-STD-1474	- Noise Limits for Army Materiel.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the soliciation.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations 40 CFR and the Federal Register and should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards
ASTM Designation: A525 -- Specifications for General Requirements for Delivery of Zinc-Coated (Galvanized) Iron or Steel Sheet, Coils and Cut Lengths Coated by the Hot-Dip Method.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SAE, INC.
SAE Standards and Recommended Practices
J198 - Windshield Wiper Systems - Trucks, Buses, and Multipurpose Vehicles.
J381 - Windshield Defrosting Systems Test Procedure - Trucks, Buses, and Multipurpose Vehicles.
J382 - Windshield Defrosting Systems Performance Requirements - Trucks, Buses, and Multipurpose Vehicles.
J537 - Storage Batteries.

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	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
*	J683	- Tire Chain Clearance - Trucks, Buses and Combinations of Vehicles (DoD adopted).
	J688	- Truck Ability Prediction Procedure (DoD adopted).
	J816	- Engine Test Code - Spark Ignition and Diesel.
	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA
Vehicle Code of California

(Application for copies of State of California publications should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standard), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1.1 through 3.1.1.10, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's

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technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. Treatment, painting, identification marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2). When specified (see 6.2), the exterior color shall be gloss yellow, matching color chip No. 13432 of FED-STD-595. When specified (see 6.2), concealed markings shall be furnished.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle chassis and underbody, including fenders, skirts, and wheelwells, shall be rustproofed in accordance with FED-STD-297.

3.1.1.3 Heavy duty cooling system. A heavy duty cooling system shall be furnished that will maintain the coolant at a temperature below the boiling point, with vehicle loaded to the rated gross vehicle weight (GVW) and operated at 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.1.1.4 Coolant system and indicators. The coolant system shall include a de-aeration system, a surge tank, or a coolant recovery reservoir of not less than 2-quart capacity. A low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

3.1.1.5 Engine provisions for bus heating. The engine shall have plugged openings for the purpose of supplying hot water for the bus heating system. The openings shall be suitable for attaching a 3/4-inch pipe thread hose connector. The engine shall be capable of supplying water having a temperature of at least 170°F at 0°F ambient temperature at a flow rate of 6 gallons per minute at the return end of 30 feet of one-inch inside diameter (i.d.) automotive hot water heater hose.

3.1.1.6 Wood treatment. Wood for the floor (see 3.5.3) shall be treated in accordance with MIL-STD-1223.

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3.1.1.7 Rear towing devices. Two closed loops for towing the bus shall be provided on the rear of the bus. The towing loops and the rear of the bus shall withstand, without any signs of failure, lifting the rear end of the bus, turning the bus within its minimum turning radius, and backing the bus on a level roadway. Decals shall be provided on front and rear bumpers stating "LIFT AND TOW FROM REAR OF BUS." Complete towing instructions shall be furnished with the operator's manual.

3.1.1.8 Front tow attachments. Two tow chain attachment devices shall be provided at the front of the bus to allow recovery of the bus from a ditch or other impediment. Each front attachment device shall provide an ultimate strength at least equal to the GVW rating of the vehicle furnished.

3.1.1.9 Towing brakes (type III). When specified (see 6.2), in addition to the brake requirements specified herein (see 3.4.11), an emergency system for controlling at least the front brakes of the type III, 44-passenger bus from a towing vehicle shall be provided. The installation shall be complete with air brake couplers; a relay emergency valve with no-bleed-back feature, except when a spring applied emergency brake is furnished; and additional air lines and fittings. The towing brake installation shall not compromise conformance to any Federal Motor Carrier Safety Regulation or Federal Motor Vehicle Safety Standard referenced herein. Couplers shall be mounted in a protected location at the rear of the bus and shall permit ready attachment of air hoses from the towing vehicle. The service coupler shall be located on the curbside and an emergency coupler on the roadside.

3.1.1.10 Ambulance features. Ambulance conversion features shall be provided as specified herein for type III and IV buses.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. Vehicles and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture, including standards applicable only to school buses, except:

(a) Exterior color shall be as specified in 3.1.1.1
(b) School bus signal lamps may be omitted when specified (see 3.4.2.6.4)
(c) Compliance with FMVSS No. 222 is not required. Seats shall be in accordance with 3.5.8 in lieu of FMVSS No. 222.

3.2.2 Air pollution control. Vehicles shall comply with the Environmental Protection Agency Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in

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effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level at the operator's location with the driver in the seated position shall be not greater than 85 dB(A) when measured in accordance with MIL-STD-1474. The vehicle exterior sound level shall conform to the Interstate Motor Carrier Noise Emission Standards of the Environmental Protection Agency when tested in accordance with the regulations of the Department of Transportation, Part 325. When specified (see 6.2), the exterior sound level shall not exceed the noise limits established by the California Vehicle Code, Section 27200.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and body with all attachments, accessories, and equipment; a full complement of fuel, lubricants, and coolant; and the operator weight (computed at 175 pounds).

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall include the curb weight and a payload evenly distributed throughout the passenger seat and the standee area. The GVW shall be not less than 26,000 pounds for 30-passenger buses and not less than 29,500 pounds for 44-passenger buses. In addition, the payload allowance used to determine the GVW shall be not less than 6,000 pounds for 30-passenger buses and not less than 8,800 pounds for 44-passenger buses.

3.2.6 Dimensions. The overall bus dimensions shall be as follows:

(a) Overall width shall be not greater than 96 inches. Width over tires shall be not more than 100 inches;
(b) Overall height shall be not greater than 125 inches. When air conditioning is furnished, the overall height shall be not greater than 140 inches;
(c) Overall length shall be not greater than 36 feet for 30-passenger buses and not greater than 38 feet for 44-passenger buses, including all front and rear protrusions;
(d) Ground clearance, measured with the vehicle loaded to the GVW, specified in 3.2.5, shall be not less than 8-1/2 inches under axles and not less than 10 inches under all other parts.

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity.

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3.2.8 Accessibility. The design of the vehicle and optional equipment shall permit ready accessibility for routine servicing and shall permit accessibility for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the bus loaded to its specified GVW.

3.3.1.1 High speed gradeability. At 55 miles per hour (mph), the vehicle shall ascend a grade of not less than 1.2 percent for type III buses and 1.5 percent for type IV buses. Gradeability requirements shall be met with the transmission in direct drive and, when a multispeed axle is furnished, with the axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, and shall provide a vehicle speed not greater than 4 mph.

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. Under all conditions of loading, the service brakes shall stop the bus, from a speed of 20 mph, within a distance of 35 feet from the point at which movement of the brake pedal begins. Deceleration and stopping distance shall be demonstrated on dry, hard, approximately level road surface that is free from loose material.

3.3.3 Turning radius. The turning radius, defined as the distance from the turning center to the centerline of the tire describing the largest circle while the vehicle is executing its sharpest practicable turn, shall be not greater than 34 feet.

3.4 Chassis.

3.4.1 Engine. The engine furnished shall be the manufacturer's standard or optional diesel engine for the commercial model vehicle which meets or exceeds the requirements of this specification. The engine shall be mounted at the front end of the vehicle or mid-mounted under the vehicle floor between the front and rear axles.

3.4.1.1 Diesel. The diesel engine shall be liquid cooled, compression ignition, two-stroke or four-stroke cycle, with not less than six cylinders. The engine assembly shall have been in fleet usage

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for at least one year prior to the contract award. The engine furnished shall be as recommended by the engine manufacturer for the vehicle model furnished. The engine net torque rating shall be not less than 414 pound-feet. The engine net torque rating and net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349.

3.4.1.2 Engine compartment. Engine compartment ventilation and design shall allow operation without damage of engine, transmission and all components when the vehicle is operated in ambient temperatures of up to 120°F. Ready access to the engine compartment shall be provided for servicing and routine maintenance of engine and engine components. Hinged engine compartment covers or doors shall be provided with a device to hold them open for servicing and maintenance.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. Except for vehicles having an automatic transmission, an engine governor shall be furnished and set to limit the engine speed to the maximum recommended operating speed. The governor shall be sealed at the specified setting.

* 3.4.1.5 Fan clutch. A fan clutch shall be furnished. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.6 Coolant temperature control. Thermostatic control of the engine coolant temperature shall be provided to maintain the engine temperature within the engine manufacturer's recommended operating range. The control shall include complete thermostatic control of all coolant flow through the radiator, or partial thermostatic control of the coolant flow through the radiator and thermostatically controlled radiator shutters. When power plant heaters are specified (see 3.4.1.7), thermostatically controlled radiator shutters shall be provided.

3.4.1.7 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, engine oil heater, and fuel warmer shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac) and shall be wired through a junction block to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the

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vehicle and a standard, weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperature (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with the vehicle component operation and without loose excess wire. A carrier for the connecting cable shall be mounted within the engine compartment or stowage compartment(s) and shall provide positive cable retention during vehicle operation. Heaters shall be furnished as follows:

(a) Coolant heater, 1,500-watt minimum rating shall be installed in the engine block or lower coolant inlet hose. An engine thermostat with an operating range of 170°F to 195°F shall be installed;

(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening;

(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F;

(d) A fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.9 Fuel fired engine preheater. When specified (see 6.2), a diesel fuel fired engine water heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat and circulating pump; and shall be connected directly across the engine coolant inlet and outlet. The heater shall be capable of starting and operating at minus 60°F and shall heat the engine to plus 40°F from minus 60°F in not more than 1 hour. The system shall be equipped with a light, visible to the driver, to indicate that the preheater is operating.

3.4.2 Electrical systems. The electrical systems shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

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* 3.4.2.1 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

* 3.4.2.2 Starting system. A 12- or 24-volt direct current (dc) starting system with a 12-volt lighting system shall be furnished. Engine starting equipment shall include an ether starting system or a glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.3 Alternator. The rated amperage capacity of the alternator provided shall be not less than 130 amperes at not more than 60 percent of the engine governed speed and not less than 50 amperes at engine idle speed. A transistor type voltage regulator shall be furnished.

3.4.2.4 Safety switch. An engine starting device and a safety switch shall be provided in the engine compartment on buses with mid-mount engines so arranged that the engine cannot be cranked from the front end with the switch in the "off" position. Means shall be provided to prohibit cranking the engine using the engine compartment starting device except when the transmission is in the neutral position.

* 3.4.2.5 Batteries. Batteries shall be of 12-volt potential. The total reserve capacity rating measured in accordance with SAE J537 shall be not less than 480 minutes. The total cold cranking rating at 0°F, measured in accordance with SAE J537, shall be not less than 1,740 amperes. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131. The batteries shall not be exposed to excessive engine or exhaust heat. Batteries shall be mounted on a sliding tray with rollers in an enclosed compartment. The sliding tray shall support the batteries in the pulled out position and, in this position, shall provide for ease of battery inspection, maintenance and replacement. The tray shall be coated with acid resistant material. Battery access door shall be provided on the side of the body.

3.4.2.6 Lighting. All vehicle (chassis and body) lights, reflectors, and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and

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393.22 through 393.26. The electrical system shall be 12 volts. Lights and reflectors shall not be mounted on rub rails (unless recessed and fully protected) or vehicle bumpers. An illuminated rear license plate holder shall be provided. The bus shall be equipped with the manufacturer's standard foglights. Two backup lights shall be provided at the rear of the vehicle. A switch to turn the backup lights on and off shall be provided inside the rear door.

3.4.2.6.1 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The vehicle shall be provided with single-faced front signal units and with single-faced rear signal units installed in accordance with SAE J588. Operating units shall be provided with a visible and audible flash indicator. Turn signals shall be of the self-cancelling type. Additional armored type amber turn signals, one mounted on each side of the bus above or near the front wheel well, shall be furnished.

3.4.2.6.2 Passenger signal buzzer. When specified (see 6.2), the manufacturer's standard signal buzzer equipment shall be furnished, operated by suitable pull cords on both sides of the bus, and provided with a driver's cutout switch.

* 3.4.2.6.3 Interior lighting. The interior lighting shall include 15-candlepower domelights of the grade and quality providing not less than 12 square inches of polished, reflective surface fitted with a frosted translucent lens providing not less than 30 square inches area. For type III, 44-passenger buses, domelights shall be installed over each row of two-passenger seats, placed over the first or second seat from the front, and thereafter over each second seat. For type IV, 30-passenger buses with perimeter seats, domelights shall be placed over the aisle side of each first or second seating position from the front, and thereafter over the aisle side of each second seating position. One domelight shall be installed over the driver's area and controlled with a separate switch. Fluorescent lamps and fixtures may be furnished in lieu of the domelights. Lamps shall be located on the longitudinal centerline of the aisle of the passenger area for the entire length. There shall be not less than 4 lamp fixtures in the row. Lamps shall be not less than 40 watts, and of the high output capacity type. Lamps shall provide a broad distribution of light. Lamps shall be white. A nonglare reflector or shade and tube guard shall be provided. The fluorescent lamps shall be furnished with inductive-capacitive type filters for each lamp to reduce radio interference (see 3.4.2.1). Door-operated ceiling lamp(s) and step lamp(s) shall be furnished at the service door and installed to provide illumination of the stepwell

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and ground adjacent to the steps. An emergency door identification red light with not less than a 3-inch diameter lens shall be provided in accordance with Federal Motor Carrier Safety Regulation 393.92 (see 3.5.5.2). An audible signal to the driver shall be provided to indicate when rear and emergency door(s) are not properly secured during operation of the vehicle.

3.4.2.6.4 School bus red signal lamps, (type III). Unless otherwise specified (see 6.2), the type III, 44-passenger vehicle, shall be equipped with four red signal lamps conforming to Federal Motor Vehicle Safety Standard No. 108.

3.4.2.6.5 Floodlight. A center roof-mounted floodlight shall be furnished over the rear door(s) with the operating switch located inside the rear door(s). The floodlight shall be adjustable and shall include a locking device to secure the floodlight in any selected position. Mounting shall permit directing the light beam to all points two feet or more to the rear of the vehicle. The switch shall be labeled as to function.

3.4.2.6.6 Interior lighting (type IV). Interior lighting for type IV vehicles shall be as specified in 3.4.2.6.3. In addition, with the exception of the door operated ceiling lamp(s) and step lamp(s) and the red emergency lights, the intensity shall be capable of being controlled by the driver. In addition, a domelight with an attendant-controlled on and off switch shall be furnished over each stretcher section approximately 36 inches in back of padded barriers (modesty panels) and 12 inches from the wall. These domelights shall be provided with a translucent lens and each light shall provide not less than 40 foot-candles of illumination at seat level.

3.4.2.7 Master cut-off switch. When specified (see 6.2), an ignition key or driver manually operated master switch shall be provided to control a magnetic solenoid to shut off all body electrical equipment except: (a) headlights, (b) running lights required by FMVSS 108 and (c) the starting motor. The switch shall be identified as to function. The solenoid shall be rated for operation under the continuous full load.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. A single-stage or two-stage dry type air cleaner shall be furnished. The air cleaner shall include a progressive

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type indicator to display the degree of air restriction. The indicator shall be located adjacent to the air cleaner and shall have a red line or other clear mark to indicate when the filter should be changed.

* 3.4.3.2 Fuel tank(s). Safety fuel tank(s) conforming to Federal Motor Carrier Safety Regulations 393.65 and 393.67 shall be furnished. Total capacity of the fuel tank(s) shall be not less than 60 gallons for vehicles without air conditioning and not less than 80 gallons for vehicles with air conditioning. When more than one tank is furnished to feed the main diesel engine, means shall be provided to assure an equalized fuel level between the tanks. The tank(s) shall be located under the floor of the bus. A filler shall be arranged for safe and convenient filling from the exterior body. The access for the filler neck shall be designed to prevent the entry of spilled fuel to any point inside the passenger compartment or entrapment between body walls. Sealing may be accomplished by use of a neoprene rubber thimble or grommet, or other device installed in the access hole. When the tank filler cap is located behind body skirting, a front hinged access door shall be furnished. A photoetched, corrosion-resistant plate, permanently protected decal or stencil, shall be located in a conspicuous place near the filler opening and shall state "diesel fuel" and "main engine" or "main and auxiliary engine" as appropriate. When an air conditioning system is furnished (see 3.5.11), the air conditioner auxiliary diesel engine may operate from the main chassis fuel tank(s) or from a separate tank specified in 3.4.3.2.1. When specified (see 6.2), when the air conditioner engine operates from the main engine fuel tank(s), means shall be provided to assure the auxiliary engine does not draw below 20 percent of the chassis fuel tank capacity.

* 3.4.3.2.1 Air conditioner auxiliary engine fuel tank. When an air conditioning system is specified (see 3.5.11), and a separate fuel tank to supply the air conditioner auxiliary engine is furnished, the auxiliary fuel tank shall have a capacity of not less than 20 gallons and shall be located under the floor. The auxiliary tank shall be a safety fuel tank of welded seam construction. The tank shall be provided with a fuel level sensor, connected to a separate fuel gage visible to the seated driver. Gage shall be identified as air conditioner fuel. A drain shall be provided in a fuel tank sump, located to avoid drainage of fuel on any part of the vehicle. A fuel tank filler neck shall be located on the side of the bus with access through a flush type door. Filler opening shall be identified as specified in 3.4.3.2 and shall indicate "Diesel Fuel - For Air Conditioner Engine Unit Only." A shut off valve shall be provided in the fuel line adjacent to the tank. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

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* 3.4.4 Exhaust system. An exhaust system, conforming to Federal Motor Carrier Safety Regulation 393.83, shall be substantially constructed, effectively leakproof, and quiet in operation. A properly supported exhaust tailpipe shall be located to properly and effectively minimize the possibility of exhaust fumes entering the passenger area under all operating conditions. The exhaust system shall include muffler(s).

* 3.4.5 Transmission. The transmission input torque capacity shall be at least equal to the maximum torque delivered by the engine. Transmission and axle gear ratio shall be selected to provide the performance specified in 3.3.1 through 3.3.1.2.

3.4.5.1 Automatic or semiautomatic transmission. Unless otherwise specified (see 3.4.5.2), an automatic or semiautomatic transmission shall be provided. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. The normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. When specified (see 6.2), the automatic transmission shall have a lockup in the two highest forward speeds.

3.4.5.2 Manual transmission. When specified (see 6.2), the manufacturer's standard manual synchromesh transmission shall be furnished. The transmission shall provide one reverse speed and not less than five forward speeds.

3.4.5.2.1 Clutch. The clutch shall be the largest capacity clutch offered for the size of the engine furnished, with clutch torque capacity exceeding the maximum delivered engine torque.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction. Propeller shaft guards shall be provided in accordance with Federal Motor Carrier Safety Regulation 393.89.

3.4.7 Chassis frame. The chassis components shall be forward driver control, heavy duty type with a frame for bus body mounting or a unit body and frame construction. The frame shall be continuous from bumper to bumper or shall be continuous and of the same cross-section at least from the rear bumper to the forward front suspension hanger bracket. When continuous frame rails are furnished, the section modulus and the resisting bending moment of the frame rails shall be as

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recommended by the chassis and body builder(s) to support the body and specified load under severe operating conditions. When a noncontinuous frame rail is provided, the front rail section shall be reversed to provide clearance for mounting a diesel engine between the frame rails; the front and rear rail sections shall be overlapped and joined together by both bolted and welded construction; and the resisting bending moment shall be not less than 640,000 inch-pounds for the main rail section and not less than 390,000 inch-pounds for the front rail section. The bus chassis shall be built by the bus body builder or to the bus body builder's specifications for the specific model being furnished. The chassis frame or unit body and frame and undercarriage shall be so designed and constructed as to support body and load and to maintain stability under severe operating conditions.

* 3.4.8 Suspension. The vehicle shall be equipped with the bus manufacturer's standard or optional system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to its specified GVW. The vehicle shall be provided with a spring, full air, or combination spring and air suspension on the front axle. Rear suspension shall be full air. The vehicle shall be provided with hydraulic double acting shock absorbers, not less than two on the front and two on the rear suspension. Suspension shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the vehicle from the applicable jacking location. Air systems shall incorporate leveling valve(s) with a time delay to avoid constant air consumption. Shock absorbers and airbags shall be fabricated of materials capable of operating in ambient temperatures from minus 50°F to plus 125°F. The suspension system shall be recommended and warranted by the chassis supplier to meet specified conditions under mobile operating conditions.

3.4.8.1 Front suspension. The spring, full air, or combination spring and air suspension design shall be such that the natural frequency of vertical vibration at the front wheels under all conditions of loading shall not exceed 2.0 cycles per second. Torque rods shall be provided if available as manufacturer's standard or option to increase longitudinal and lateral stability.

* 3.4.8.2 Full air rear suspension. The rear axle equipped with full air suspension shall have two or four air springs of adequate diameter. The suspension design shall be such that the natural frequency of vertical vibration about any transverse axis shall not exceed 1.5 cycles per second at rated capacity for the specified GVW. The full air system shall have none of the spring weight carried on anything other than air springs. No other spring medium such as leaf, coil, or rubber springs may be used in conjunction with the air springs. The two-air-spring style system shall include shock absorbers; two

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one-piece beams and two air springs to transfer the load directly to the axle with the forward portion of the one-piece beam free to move in low-friction wear pads; and a height-control valve system capable of maintaining a given frame height. The full air system shall incorporate four radius rods for the four-air-spring type and three radius rods for the two-air-spring type to control lateral and longitudinal movement. Radius rods shall transmit driving and braking forces to the chassis and body. Each end of the radius rods shall be equipped with rubber bushings that do not require periodic lubrication. The suspension shall be so designed that the vehicle can be lifted and towed from the rear without damage to the suspension.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when the vehicle is loaded to its specified GVW. The vehicle shall be equipped with single front and single rear axles. Unless a two-speed axle is furnished, a single reduction drive axle shall be provided at the rear. Torque rods shall be furnished for the rear axle(s) if available as the manufacturer's standard or option. Axle gear ratio furnished shall enable vehicle performance required in 3.3.1 through 3.3.1.2.

3.4.9.1 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.9.2 Two-speed axle. When specified (see 6.2), a two-speed axle shall be furnished. The axle shall be equipped with an electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. The gear ratio shall provide the performance specified in 3.3.1 through 3.3.1.2.

3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rim size, tire size, type and ply rating shall be the same for all wheels on each vehicle. Tire and rim ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. When specified (see 6.2), disc type wheels shall be furnished.

3.4.10.1 Tires. Tires shall be tube or tubeless type with highway tread. Steel belted radials, or when specified (see 6.2) bias ply tires, shall be furnished. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the

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ground, with the vehicle loaded to its specified GVW. Tires shall conform to the Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Inner tubes. For tube type tires, inner tubes shall be of the heavy duty type, and shall be of the proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Balancing. Each tire shall balance within practicable limits. Wheels, hubs and brake drums shall be effectively in balance. Balancing shall be adequate to preclude wheel shimmy at all vehicle speeds.

3.4.10.4 Spare wheel assembly. When specified (see 6.2), a spare wheel or rim shall be mounted on a winch type carrier and installed in a readily accessible location. The spare wheel assembly shall be so located as not to protrude beyond the body. The carrier design shall enable removal or mounting of a spare wheel assembly using only the tools specified in 3.4.16. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material.

3.4.10.5 Spare tire and wheel assembly. When specified (see 6.2), a spare tire and wheel assembly shall be furnished. The assembly shall include the winch type carrier, the spare wheel or rim specified in 3.4.10.4 and a mounted and inflated spare tire. Moving parts and threaded fasteners used to secure the spare tire in the carrier shall be constructed of or plated with corrosion-resistant material. The spare tire shall be of the same size, type, tread design and load range (ply rating) as the tires furnished on the vehicle.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42, and 393.44 through 393.52.

3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-header type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm)

* (b) Air storage reservoir(s) with a total capacity of not less than 3,600 cubic inches; each tank shall be equipped with a drain, and with safety and check valves between the compressor and the last reservoir tank

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(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
(i) Relay emergency valve
(j) Automatic slack adjusters.

* 3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.12 Horn. An electric horn shall be provided. In addition, when specified (see 6.2), an air operated horn shall be provided.

3.4.13 Steering. The manufacturer's standard power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual, air operated windshield wipers. The wiper system furnished shall conform to SAE J198 and shall sweep the maximum practicable area of windshield surface. Windshield washers with control on instrument panel shall be provided.

3.4.15 Bumpers. A channel type front bumper shall be provided. The rear bumper shall conform to Federal Motor Carrier Safety Regulation 393.86 and shall be as specified in 3.5.12.6. The top of the front bumper shall be provided with a skid-resistant finish to assist personnel in windshield cleaning (see 3.4.18).

3.4.16 Tools. When specified (see 6.2), the vehicle shall be furnished with the tools required for exchanging a mounted tire assembly with a spare assembly (see 3.4.10.5), and shall include at least a hydraulic jack, jack handle, and wheel nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit the removal and replacement of the wheel and tire assembly.

3.4.17 Controls and operating mechanisms. Unless otherwise specified (see 6.2), all controls, operating mechanisms, and doors shall be located for left-hand drive. Controls shall be complete and

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conveniently operable by the driver when in a seated position. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's seat. Instruments shall be panel mounted and shall be clearly visible to the driver when seated in the normal driving position. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing.

* 3.4.18 Windshield cleaning grab handles. A handle shall be furnished at each side of the windshield to provide a handhold for windshield cleaning while standing on the bumper (see 3.4.15). The handle shall be tubular metal or reinforced plastic with a tubular shape.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories and equipment furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator, or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer
(g) Recording odometer
(h) Tachometer
(i) Indicator light to show headlight high beam operating
(j) Hand throttle.

3.5 Body. The body shall be mounted on a chassis or may be of unit body and chassis construction. The bus shall be forward control type with seating for the driver forward of the front axle. The bus shall provide adequate seating for adult passengers and the litter carrying capacity specified (see 1.2), and other ambulance conversion features specified herein. Type III buses shall support 12 litters above the forward facing seats, with folding down of seat backs only. Type IV buses shall hold 6 litters on the longitudinal bench seats on each side of the vehicle and support 6 additional litters above the seats. The body shall have sides of the full-skirted type, shall present a generally symmetrical appearance, shall effectively exclude engine fumes and exhaust gases from the bus interior, and shall provide safe and comfortable transportation for passengers and driver.

3.5.1 Inside body dimensions. At the seat level, the body inside width shall be not less than 90 inches. Between the first and last vertical roof bows, through the center of the aisle, the inside height shall be not less than 76 inches.

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3.5.1.1 Front entrance clearance (front mounted engine only). The clear opening in the stepwell, in the entrance area and between the curbside modesty panel and the engine housing shall be not less than 18 inches.

3.5.2 Construction. The body and roof framing shall be constructed of steel sections, with metal panels attached outside and inside and properly insulated. All parts, where applicable, shall be fastened together in a manner which will preclude loosening of bolts, screws and rivets, and cracking of welded joints when the vehicle is properly maintained and operated under the intended service conditions. Self-tapping screws shall not be used where subjected to stress. Wood shall not be used for structural framing. Outside body panels and other sheet metal shall be of aluminum alloy or steel not less than the gages regularly furnished by the body manufacturer in buses of the type offered. The body, roof and panel joints shall be watertight and leakproof. Unless zinc coated steel panels are used, joints shall be sealed with "Alumelastic" compound or equivalent. All steel, 12 gage (0.1046 inch) and lighter, shall be zinc coated in accordance with ASTM Designation A525 and table I, Coating Class G60.

3.5.2.1 Sound level (passenger locations). The octave band sound pressure level average shall not exceed those levels shown in column A of table II. Sound readings shall be taken at three positions in the aisle: (a) next to the front passenger seat, (b) next to the center passenger seat, and (c) next to the rear passenger seat with the vehicle traveling at 30 mph in direct drive on a smooth level road with all powered accessories except the heater blower (and air conditioner, when furnished) operating and no more than four people within the vehicle. At the authorized maximum speed for the vehicle and at the noisiest position in the vehicle, the octave band sound pressure level shall not exceed those levels shown in column B of table II.

TABLE II. Acceptable noise level.

Octave band limits (Hz)	Sound pressure level dB, ref. 0.0002 microbar	
	Column A	Column B
75 - 150	84	--
150 - 300	80.5	92
300 - 600	72	89
600 - 1200	71	87
1200 - 2400	68	85
2400 - 4800	66	84

3.5.3 Floor. Floor frame and floor plates, with entrance step and wheelhousings, shall be of substantial construction, properly integrated with the body structure, and shall conform to Federal Motor Carrier

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Safety Regulation 393.84. Wood parts, including plywood, shall be treated in accordance with 3.1.1.6. Plywood used in floor panels shall be water-resistant, exterior grade fir or dense southern yellow pine. Floor panels shall be of not less than: (a) 3/4-inch, seven-ply plywood; (b) 1/2-inch, five-ply plywood over 10 gage steel; or (c) 5/8 inch, five-ply plywood over 14 gage steel. Steel channels, not less than 2 inches in depth and spaced to provide support for the loads imposed, shall be furnished under the floor.

3.5.3.1 Stepwell. The stepwell(s) shall be enclosed, with two or three steps of uniform height, and constructed of steel covered with 3/16-inch thick rubber or synthetic rubber blend ribbed material. The height of the bottom step shall not exceed 16 inches.

* 3.5.3.2 Wheelhousings. Front and rear wheelhousings shall include splash shields under the body. Splash shields shall extend from the bottom of the wheelwell to the bottom of the side skirting, front and rear. Ample clearance for the use of tire chains on at least the outside driving wheels shall be provided in accordance with SAE J683. Molded rubber splash trim shall be installed around the outside edge of both the front and rear wheelhousings.

3.5.3.3 Floor coverings. The under seat floor area and the wheelhousings shall be covered with not less than 1/8-inch thick rubber or synthetic rubber blend material; or linoleum conforming to type I, class I, grade B of LLL-F-1238. The aisle and entrance area shall be covered with not less than 3/16-inch thick rubber or synthetic rubber blend ribbed material. Coverings shall be properly installed and adequately sealed at the joints. The floor in the driver's area shall be covered with a mat or with the same materials as used in the underseat or aisle area.

3.5.3.4 Rear step/seat. When specified (see 6.2), a rear step, capable of being folded into the cargo area, shall be provided. When folded in, it shall serve as an attendant's seat that shall be not less than 18 inches deep. The step shall be not less than 15 inches wide and 10 inches deep. The step side shall have a nonskid tread. The seat side shall be padded and covered, using material identical to that used on the passenger's seats.

3.5.3.5 Rear loading ramp. When specified (see 6.2), a walk-up patient loading ramp shall be provided at the rear of bus. The ramp shall be made of lightweight materials, consistent with the specified performance criteria. The ramp shall be not less than 26 inches wide and shall have an embossed metal or fiberglass antiskid surface covered with a nonskid material. When in use, the ramp angle with the ground shall be not more than 22 degrees. The ramp shall be capable of rigidly supporting two litter bearers and one litter patient, each person

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weighing 200 pounds. The maximum deflection of the center of the ramp shall be not greater than 1-1/2 inches under any normal combination of static or dynamic loads imposed by the position or motion of the two litter bearers and one litter patient. Provisions for securely stowing the ramp when the bus is in transit shall be provided under the floor between the frame rails with access from the rear of the bus. Stowage provisions shall enable the ramp to be quickly placed in the operating position by not more than two men without the use of tools. The underfloor ramp storage compartment shall be flared out at the opening to accommodate loading the ramp. When stowed, the ramp shall not interfere with the use of the attendant's seat (see 3.5.3.4), or interfere with the opening and closing of the rear door.

3.5.4 Sides and roof. The sides and roof shall have sheet metal outer panels, lined on the inside and integrated with the body structure.

3.5.4.1 Exterior. The sides shall be fullskirted and equipped with two full-length rub rails extending along each side of the body; one below platform level and one at or approximately at seat height. A substantial snow rail at the bottom of the skirt will be acceptable in lieu of the lower rub rail, if such rub rail is not offered by the manufacturer as a standard or optional item.

3.5.4.2 Interior. The body interior, including sides, roof, and front and rear ends, shall be fully lined with sheet metal panels.

3.5.4.3 Insulation. Nonsettling, nonhygroscopic, fungus resistant, fireproof insulation shall be installed in the roof, side, and end walls. In addition, a coat of material shall be applied to the inner (not exposed) surfaces of exterior panels and other exterior sheet metal as added insulation and to prevent drumming. Insulation shall be of sufficient density to enable the air conditioner to meet the requirements of 3.5.11.

3.5.4.4 Destination sign compartment. When specified (see 6.2), a destination sign compartment shall be installed at the front center of the roof above the windshield. The compartment shall provide not less than 180 square inches of glass enclosed frontal area of a height to permit reading of 6-inch letters on the sign roller and shall be readily accessible through a latched door from inside the bus. The destination sign shall be a roller type of a capacity to apply 15 listings of 6-inch high letters and shall be operable from the interior of the bus. Unless otherwise specified (see 6.2), a blank destination sign roll shall be furnished. The compartment shall be illuminated to permit reading of the destination sign at night.

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3.5.5 Doors. A service door shall be provided at the front, forward of the front axle, on the side opposite the driving controls, for use by entering and departing passengers. Rear door(s) shall be furnished at the rear for the loading of litter patients. When specified (see 6.2), a hasp for padlocking the service door and a locking bolt for the emergency door(s) shall be provided. The locking bolt for the emergency door(s) shall include an interlock that prevents the engine from starting when the emergency door(s) is locked. Doors shall be provided with compression or overlapping seals to minimize entry of dust, cold air, rain, and road wash.

3.5.5.1 Service door. The service door shall be of the folding leaf type or of one-piece type with a safety glass panel in the upper and lower sections. The door shall have manual or air actuated controls conveniently located for driver's operation. When specified (see 6.2), the door shall be air actuated with individual remote controls for driver's operation provided for each door furnished. Operating linkage shall be installed within the body or enclosed to preclude accumulation of mud, ice and snow. Minimum dimensions of clear opening shall be 24 by 74 inches, unobstructed by door seals or door opening mechanism.

3.5.5.2 Emergency door. The emergency door shall be of the sedan type, equipped with a slide bar, a cam operated latch, and with a readily operable interior handle. A permanently installed handle shall be provided for opening the emergency door from outside the bus. A switch operated by the door bolt shall be connected to a red warning light on the instrument panel to indicate to the driver when the emergency door is not safely latched. A red emergency door identification light (see 3.4.2.6.3) shall be installed on the emergency door header. The red warning light shall actuate when the ignition switch is in the "on" position. The header shall be marked with the following block letters not less than 2 inches high: "EMERGENCY EXIT" or "EMERGENCY DOOR." Rear door(s) for the buses may be furnished in lieu of an emergency door if these door(s) meet the Federal Motor Vehicle Safety Standards and the requirements specified in this paragraph.

* 3.5.5.3 Rear door(s). A rear door of the sedan type or two rear doors, one on each side of center, providing a clear opening width of not less than 49 inches and a height of not less than 64 inches, shall be furnished for loading and unloading litter patients. Door(s) shall be hinged at the outer edges and braced to prevent sagging; shall open not less than 180 degrees; shall be furnished with safety glass windows in the upper section; and shall be provided with safety catches to prevent inadvertent closing. Padding shall be installed above the door opening to prevent head injury. Door(s) shall be equipped with a latch operable from inside and outside the body. Further, rubber pads or other means of protection shall be installed at a location that will

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prevent damage to the body side(s) and door(s) when door(s) is (are) in the full open position. A red reflector of 3-inch minimum diameter shall be installed on the interior surface of each rear door. The reflectors shall be positioned to provide maximum visibility when the doors are in full open position. At the manufacturer's option, reflective tape, dark red, conforming to L-S-300, type I, class 3, reflectivity I, may be furnished. The tape shall be 4 inches wide and shall extend the full width of the door panel(s).

3.5.6 Windshield. A windshield providing maximum practicable driver vision shall be installed at a slope sufficient to minimize external glare or reflections from inside the body.

3.5.7 Windows. Windows shall be of the size and arrangement suitable for the intended service and shall provide maximum practicable vision. Safety glass shall be used in all windows. Windows shall be of the horizontal sliding type with anodized aluminum frames. Windows shall be provided with fingertip control permitting ready opening and adjustment to various desired openings. Windows shall be rattleproof and, when closed, rainproof, waterproof and windproof. All windows shall be fitted with locking devices. Driver's window(s) shall permit use for signaling and ventilation.

3.5.7.1 Horizontal sliding windows. In addition to the requirements of 3.5.7, horizontal sliding type windows shall be of the push-out type, hinged at the top, and of adequate size to permit unloading litter patients through the windows in an emergency. Stops to restrict the horizontal sliding action of the windows shall be furnished and shall be set to provide an opening of approximately (but not exceeding) six inches.

3.5.7.2 Tinting. When air conditioning is furnished (see 3.5.11), tinted, heat absorbing window safety glass shall be furnished. Entrance door windows, driver's left side window(s), and the windshield shall be in accordance with the manufacturer's standard for tinted glass. Side windows in the passenger area shall be tinted to double density allowing approximately 50 percent light transmission. Rear window(s) shall be tinted to quadruple density allowing approximately 35 percent light transmission.

3.5.8 Seating. Seating shall comply with Federal Motor Carrier Safety Regulation 393.91. Seats shall provide capacity for the required number of adult passengers (see 1.2). Seats located at or near wheelhousings shall be installed to minimize interference with passenger comfort. Seats shall be secured to prevent shifting under all operating conditions. Seat frames shall be of formed, metal tube construction. All bolts, nuts, and fasteners shall be of corrosion-resistant material. On type III buses, all 2-passenger seats being within a row may be mounted on a recessed track to permit varied arrangement of seats.

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3.5.8.1 Seating for type III. Seats shall be the manufacturer's standard 34-inch width. Width of rearmost seat(s) and front seat at entrance may be reduced to a nominal 26 inches, if required, to provide access to entrance and emergency exit. Seat cushions shall have a free depth of not less than 17 inches from the front edge to the seat back. Seat cushions shall be solid foam padding tapering from a minimum of 5 inches at the front to 3 inches at the rear or not less than 2 inches of padding on a spring base. Seat backs shall be shaped and padded to assure passenger comfort. Seat backs shall be not less than 18 inches free height from cushion to the top of the seat back cushion. The top, the upper corners, and the upper 10 inches of the rear surface of each passenger seat back (except the rearmost seats) shall be constructed of or covered with force distributing material. The material shall minimize force and spread the area of contact upon impact by an occupant in the next seat to the rear. The rear surface of the seat back below the force distributing material shall be upholstered or covered with aluminized steel or bright finish zinc coated steel. Seat assemblies shall be installed on not less than 29-inch centers. Where padded barriers or obstructions are located at door entrance, the front of the seat cushion shall be not closer than 10 inches from the padded barrier or obstruction. Seating shall provide two rows of two-passenger seats, all facing forward. The seatbacks shall be adjustable without the use of tools, and shall fold down flat on the seat cushion in an essentially horizontal position. With seat backs folded down, sufficient space shall be provided above the seats to allow both tiers of loaded litters to be put in place (see 3.5.13). All passenger seats shall be readily removable from the inside of the bus, without the use of tools. Seats may be mounted in recessed track to facilitate removal and reinstallation. Seat track and mounting devices shall not present a tripping hazard to litter bearers during litter loading and unloading operations.

* 3.5.8.2 Seating for type IV. The type IV bus seating shall be of longitudinal bench type installed on both sides of the vehicle. Each bench shall be divided into 3 sections by 8-inch high dividers of heavy gage corrosion-resistant metal. Exposed edges shall be covered with stainless steel, stainless steel clad, or plastic coated steel tubing at least 3/4 inch in diameter. Each section shall be a minimum of 94 inches in length, 24-1/4 inches in width, and 16 inches in height from the floor. Each section shall support 5 seated passengers or a NATO, Army, or Stokes litter. Four 5x3-inch (with 1/2-inch lip) litter leg cups of corrosion-resistant metal shall be mounted in parallel pairs on each section of the seat. The seat cushion shall be formed around the litter leg cup, leaving the cup exposed. The centerline of the litter leg cups will be approximately 24 inches from the dividers, 3 inches from the wall on 20 3/4-inch centers crosswise, and 46-inch centers lengthwise (see figure 1).

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3.5.8.2.1 Seat and backrest for type IV. For type IV vehicles, both the backrests and the full seat around the litter leg cups shall be padded with flexible polyethylene foam. The foam shall be nonallergenic, flame-retardant, nontoxic, and shall not absorb moisture or body odors. Density shall be 2 pounds per cubic foot to 2.4 pounds per cubic foot. Firmness of the foam shall be 35 pounds to 70 pounds at a 25 percent deflection over an area of 50 square inches. The foam padding shall be capable of withstanding 100 percent humidity and an ambient temperature from minus 65°F to plus 125°F without any evidence of deformation or cell breakdown. Flexible latex polyester foam padding is not acceptable. Thickness of padding shall be 1-1/2 inches for the seat and 2 inches for the backrests. Padding for seats and backrests shall be covered with material specified in 3.5.8.3. The height from the top of the leg cup lip to the top of the upholstery shall be not greater than 1 inch. The backrests shall run the full length of the seat bench. The lower edge shall be 8 inches above the seat and shall extend to the lower edge of the windows. The seats and backrests shall be designed so as not to require removal or hamper the effectiveness of the vehicle when utilized to transport litter patients.

3.5.8.2.2 Safety straps for type IV. Each bench seat section of the type IV vehicle shall be provided with three 2-inch wide nylon safety straps with proper anchors so as to strap in a litter patient at the chest, waist, and knees when an Army litter is placed on the bench seat. Litters are loaded head forward. Each strap shall be not less than 72 inches in length. Two additional straps shall be provided to prevent movement of the litter. It is intended that the straps be applied to the exposed litter pole, nearest the vehicle wall, at litter head and foot. All of the safety straps and buckles shall be of the automotive seat belt type and quality. Safety strap attachment hardware shall preclude undue wear of the strap webbing at the point of attachment. All safety straps shall be of sufficient length for the stipulated intended usage, and shall be provided with means for rapid adjustment and neat storage when not in use.

3.5.8.3 Seat cover. Seat and force distributing material for type III and IV vehicles shall be first-grade, heavy duty, plastic coated, grained-surface fabric of not less than a 42-ounce weight per linear yard of 54-inch width. All seat materials shall be flashproof, flameproof and nonexplosive. The color of the upholstery shall be appropriate for the exterior vehicle colors and shall be compatible with interior colors.

* 3.5.8.4 Driver's seat. The driver's seat shall be heavy the duty bus-type, pedestal-mounted and adjustable fore and aft, and for height without the use of tools. When a front mounted engine is furnished, the driver's seat shall also be adjustable sideways and with the seat in the extreme left side slide position there shall be a clear pass through of

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not less than 7-1/2 inches between the seat and engine housing. The seat shall be positioned for convenient and safe operation of all controls by the driver. The seat cushion and back shall be padded with full-depth foam rubber, contoured to provide maximum driver comfort, and covered with heavy duty vinyl simulated leather or polyester material. A lap style seat belt with automatic locking retractors shall be installed for the driver's seat.

3.5.9 Stanchions and grab rails. One padded stanchion shall be located near the driver with a header rail and a slide or roll type opaque curtain and a padded cross rail to protect the driver from passenger interference. A padded stanchion or padded grab rail with a modesty panel (type III bus only) shall be provided at the rear of the service door entrance. A stanchion or grab rail at the front of service doors, arranged to safely assist passengers entering or leaving the bus, shall be provided. Full length grab rails shall be attached to the ceiling on each side of the aisle extending from the face of the first two-passenger seat to a point within 18 inches of the face of the rear seats. The stanchion or grab rail, header rail, and cross rail located behind the driver and at the service door shall be of stainless steel, stainless steel clad, or plastic-coated steel tubing, at least one inch in diameter.

3.5.10 Heating, defrosting and ventilating. The heating, defrosting and ventilating system shall conform to Federal Motor Carrier Safety Regulations 393.77 and 393.79.

3.5.10.1 Heater. A heating system shall be furnished capable of maintaining the temperature inside the bus at a minimum of 70°F midway between the floor and ceiling, in an ambient air temperature of 0°F. The bus heating system shall also be capable of raising the interior temperature of the bus from 0°F to 50°F in 20 minutes (with the bus soaked in an ambient air temperature of 0°F for 15 hours and with no more than a 10 minute engine warmup time). Temperature rise shall be measured 36 inches inward from the side walls, windshield and rear walls. Heat shall be evenly distributed in the aisle area. The system shall also be capable of providing 20 percent fresh air and 80 percent recirculated air or 100 percent recirculated air with controls for proportioning the amount of fresh air from 0 to 20 percent. Blower capacities shall provide a ventilating airflow of at least 1,500 cubic feet per minute (cfm). Heaters shall be located to assure flow of heated air to the driver's area, to all parts of the passenger compartment, and so that no objectionable flow is directed toward any passenger. Heater hoses shall be shielded from the passenger compartment and shall be located in protected locations. The heating system shall include not less than three heater units: one as specified in 3.5.10.2.1, one near the midpoint, and one near the rear of the bus.

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The system shall be provided with thermostatic controls. Cut-off valves shall be provided near the engine to control water inlet and outlet to heaters. A booster pump shall be provided for proper water circulation.

3.5.10.1.1 Auxiliary heater. When specified (see 6.2), and in addition to requirements of 3.5.10.1, an auxiliary heater conforming to Federal Motor Carrier Safety Regulation 393.77 shall be furnished. The heater shall operate on the same fuel as the vehicle engine and shall have a rated capacity of not less than 60,000 British thermal units per hour (Btu/hr.) The heater shall be mounted in the engine hot water system with a circulating pump of not less than 6 gallons per minute capacity. Controls for operation of the heater shall be located in the driver's area.

3.5.10.2 Defroster. Defroster outlets with suitable connections, a blower for the driver's side and a fan or blower for the right side of the windshield shall be furnished and shall be installed to provide a flow of heated air to each section or side of the windshield. The windshield defrosting systems shall conform to SAE J381 and SAE J382.

3.5.10.2.1 Windshield defroster heater. A separate heater shall be furnished to provide heated air to the driver and to the windshield defrosters. The heater shall have a capacity of not less than 30,000 Btu/hr. A manual control shall be provided to control the amount of fresh air passing through the heater. A water shut-off valve is required to permit use of the blower for power ventilation. Manual control shall be provided to adjust the flow of air to the windshield defrosters as desired.

3.5.10.3 Ventilator. Type III buses shall be provided with four power ventilators when air conditioning is not furnished (see 3.5.11). Type IV buses shall be provided with four static ventilators when air conditioning is not furnished. The ventilators shall be located in the roof, one over each wheelhousing. Outside openings for heaters and ventilators shall be so located as to minimize entry of fumes, rain, road dust, and road wash. Ventilator openings shall be provided with a weatherproof cover with means for opening and closing from inside the bus. The blower motor for power ventilators shall be capable of moving not less than 300 cubic feet of air per minute and shall have not less than two speeds controlled by separate switches mounted in a location easily accessible to the driver.

* 3.5.11 Air conditioning. Unless otherwise specified (see 6.2), the vehicle shall be equipped with the manufacturer's standard independent diesel engine driven air conditioning system conforming to 3.5.11.1 through 3.5.11.2.6. The manufacturer shall furnish to the engineering department of the procuring activity the rated capacity at

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the specified conditions of the entire unit, either by the air measurement or the flow meter method. The air conditioning system shall be powered by a diesel engine.

3.5.11.1 Performance. The air conditioner shall have a minimum net cooling capacity of 68,000 Btu/hr. in a 95°F ambient temperature and, in all conditions of solar radiation, shall be capable of maintaining not more than 80°F dry bulb and a 68°F wet bulb and an effective temperature of 74°F in the bus. The humidity shall be not less than 30 percent, and not more than 60 percent. The inside temperature shall be measured 6 inches from the floor, 6 inches from the ceiling and at seat level. Longitudinal measurement shall be taken at the front seat, at the rear seat, and midway of the passenger area. The inside temperature gradient within the bus shall be not more than 6°F. The cooling performance specified herein shall be met when the bus is traveling at 50 mph and when standing still. The air conditioning system shall be independent of the heating system and shall automatically proportion air in the ratio of 20 percent fresh air to 80 percent recirculated air. All air shall pass through the filter system. The specified air conditioner minimum net cooling capacity of 68,000 Btu shall be increased if required to meet the performance requirements specified herein.

* 3.5.11.2 Components. The air conditioning system shall be powered by an auxiliary diesel engine of sufficient horsepower so that when fully loaded at sea level conditions it shall not utilize more than 65 percent of the available power output under constant load conditions. The air conditioning power source and compressor shall be located in a compartment under the floor and shall be accessible for ease of maintenance and service. The condenser and evaporator may be mounted overhead on the roof. Air conditioner controls shall be panel mounted within easy reach of the seated driver. A second set of controls shall be located in the auxiliary engine compartment in series with the driver controls and with a safety switch to cut out driver controls. The compartment lamp switch and junction panel shall be installed in the auxiliary engine compartment. The compartment door shall be capable of being opened and secured closed without the use of tools. A thermostat control shall be located in the return air stream. The air conditioner shall be designed for bus operation and shall be complete with all necessary controls for automatic operation of the unit after the power unit has been started. The compressor shall start and stop automatically on demand of the thermostat. Mountings shall be so as to minimize the transfer of vibration. The size and weight of the assembly shall be held to a minimum consistent with the performance requirements specified herein.

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* 3.5.11.2.1 Engine components. The engine shall be a two- or four-stroke cycle diesel engine equipped with at least the following components:

(a) Self-starter, 12 volts
(b) Air cleaner.

3.5.11.2.2 Governor. A variable-speed governor shall be furnished. The governor shall automatically limit the engine speed to the selected rated speed. For type III buses, a governor adjustment feature shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

3.5.11.2.3 Electrical system. The electrical system shall utilize the bus-chassis batteries and shall conform to Federal Motor Carrier Safety Regulations 393.30 through 393.33.

3.5.11.2.4 Fuel pump. A fuel pump shall be furnished on the engine. When lifting fuel against a head of 4 feet, the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided, either as a component of the fuel pump or between the fuel pump and the carburetor.

3.5.11.2.5 Exhaust system. The exhaust system shall be furnished with a spark arresting muffler and routed to vent fumes away from the body to the side and rear of the rear wheels.

* 3.5.11.2.6 Auxiliary engine fuel tank. Fuel for the auxiliary engine shall be drawn from the main fuel tank(s) or from a separate tank specified in 3.4.3.2 and 3.4.3.2.1.

3.5.12 Body equipment. Body equipment shall be complete with all accessories listed by the body manufacturer as standard equipment and, in addition, all buses shall include the items specified in 3.5.12.1 through 3.5.12.9.

3.5.12.1 Rearview mirrors. One nonglare type mirror shall be mounted inside the bus and two mirrors outside, one each at the left and right hand sides, mounted to permit maximum vision without obstruction and to be visible on the right through the upper corner of the windshield and on the left through the driver's window(s). Outside rearview mirrors shall be of the combination (flat and convex) type. Each combination mirror shall have at least 50 square inches of flat reflective area and a convex surface having at least 20 square inches of reflective area. The inside rearview mirror shall have a reflective surface area of not less than 80 square inches.

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3.5.12.2 Sunvisor. Sunvisors shall be furnished and mounted on both windshield(s). The sunvisor color shall be compatible with interior color.

3.5.12.3 Emergency reflective triangles. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i) shall be furnished and properly stowed.

* 3.5.12.4 Fire extinguisher. A 5-pound dry chemical type fire extinguisher conforming to Federal Motor Carrier Safety Regulation 393.95(a) shall be mounted convenient to the driver. The fire extinguisher shall have an Underwriter's Laboratory rating of 10B:C or more, shall include a metal safety pin and wire, and shall conform to type I or type II, class 2, size 5 of A-A-393. The fire extinguisher shall permit visual determination as to whether it is fully charged. An energency equipment compartment with glass door or two compartments shall be furnished and located within the driver's compartment. When two compartments are furnished, one compartment shall include a bracket and space for subsequent installation of a first-aid kit, 12-unit type, properly identified. The first-aid kit size is 9-7/16 inches long, 3-7/16 inches wide, and 2-7/8 inches thick. A first-aid kit need not be furnished.

3.5.12.5 Stowage compartment(s). Provision shall be made for weatherproof stowage of tools and tire chains, when carried, to be secured by padlock. A padlock is not required.

3.5.12.6 Rear bumper. The rear bumper shall be the heavy steel channel type with anti-ride and anti-hitch features and attached in a manner to permit the bus to be pushed without permanent distortion of bumper, body, or chassis parts.

3.5.12.7 Keys. Keys in duplicate shall be furnished for all key operated locks. Keys shall be properly identified and securely packed and attached to the vehicle steering wheel or otherwise stowed in an approved manner for shipment. All locks on the same vehicle shall be of the same key code, except the ignition key.

3.5.12.8 Signs. Signs stating "DANGER - DO NOT EXTEND ARMS FROM WINDOWS" shall be affixed to the interior of the body. Signs shall be legible at a distance of 20 feet and shall be located as follows: one at the front, one at the rear, and one on each side.

3.5.12.9 Standee line. Areas prohibited to standees shall be marked in accordance with Federal Motor Carrier Safety Regulation 393.90.

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3.5.13 Litter kit (type III). Type III bus shall accommodate 12 Army and NATO pole type litters suspended in two tiers along each side of the bus. The clearance between the top and bottom rows shall be not less than 18 inches and clearance between the top row and the air conditioning system shall be not less than 19.5 inches. The litter support installation shall provide not less than 6 inches from the ends of the litters to the closed rear doors, not less than 15 inches from the nearest forward vertical obstruction, and shall have not less than 8 inches between the handle ends of the remaining litters with approximately equal spacing at each end of the central bank of litters. Installation shall secure the litters effectively without discomfort to patients. The side wall and aisle litter supports shall be furnished as specified. Each eye, hook and strap shall be capable of positively securing and supporting two litters with a 300-pound person on each litter under all dynamic operating conditions. The hook ends of hangers and wall supports shall be fitted with removable, reusable covers with sufficient padding to prevent head injury. Web straps and complicated litter holding devices will not be acceptable. Both tiers of litters shall be capable of being utilized without removal of any seat backs.

3.5.13.1 Side wall supports. Litter supports with permanently attached hooks shall be furnished on the side walls of the bus. Wall supports shall be designed to prevent litters from shifting forward or rearward. The hooks shall allow easy insertion and removal of the litters by the attendants. Sufficient space shall be provided between the edge of the hook and the wall to provide hand clearance. Hooks shall be provided with manual latching devices. The hooks shall positively secure and support 300 pounds per litter under all dynamic operating conditions. Hooks with latching devices shall manually lock closed and prevent movement of the litter out of the hook.

3.5.13.2 Aisle supports. Supports for the aisle-side of each litter shall be separate metal straps with a snap hook on the top and a combination litter-eye and hook-eye at the bottom. Eyes of "D" rings shall be provided at the ceiling of the bus to suspend the supports. An elastic type cargo line or automotive type belt strap with floor attachment shall be provided to attach to the lower hook-eye to draw down the aisle side supports and reduce vertical movement. The floor attachments shall be recessed or shall be readily removable without the use of tools.

3.5.13.3 Hanger stowage. A compartment shall be provided for the stowage of litter hangers which permits and maintains hanger stowage in an orderly manner.

3.5.13.4 Additional litter capacity. When specified (see 6.2), the type III bus shall be provided with side wall supports to retain four additional NATO pole type litters resting on the floor with seats removed. Side wall supports shall conform to 3.5.13.1.

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3.5.14 Litter support kit (type IV). The bus shall be capable of accommodating 6 litters on the seats and supporting an additional 6 Army and NATO pole type litters utilizing litter support kits. The litter support systems shall be so located to position the suspended litters directly above each bench section as indicated in figure 2. Installation shall secure the litters effectively without discomfort to the patients. Side wall and aisle litter supports shall be furnished as specified. Sufficient space shall be provided between the edge of the hook and the wall to provide hand clearance. The hooks shall positively secure and support 300 pounds per litter under all dynamic operating conditions. Hook ends of hangers and wall supports shall be fitted with removable, reusable covers with sufficient padding to prevent head injury. Web straps and complicated holding devices will not be acceptable. Side wall supports, aisle supports, and hanger stowage conforming to 3.5.13.1, 3.5.13.2 and 3.5.13.3 shall be furnished, except the platform of the aisle support straps shall have litter eyes only.

3.5.15 Body mounting. Heavy rubber cushions or antisqueak material shall be utilized between the body frame and the attaching chassis parts. Reinforcements or filler blocks shall be used where mounting devices might otherwise deform frame flanges. Mounting devices shall be locked units which will minimize loosening, but which may be tightened if necessary.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government, the contractor shall service and adjust the vehicle for operational use including at least the following: alinement of headlights; adjustment of the engine, electrical, and brake systems; filling and charging of the battery; alinement of front wheels; inflation of all tires; complete lubrication of the body, chassis, engine, and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

* 3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry. All hoses, wires and rods shall be supported and restrained where necessary to minimize vibration and to prevent damage. The supports and restraints shall not interfere with either normal or emergency operation. All sharp edges and corners shall be either padded or rounded to prevent injury to personnel. Vehicles shall be free from defects which may impair their serviceability or detract from their appearance.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The supplier shall provide the Government inspection representative instruments and such assistance as the representative may find necessary.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production vehicle of each type produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract. At the time of first inspection, operating manual(s) including procedures for removing seats, attaching litter hooks and securing loading ramps, shall be furnished.

4.3.1 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

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4.3.2 Sound, heating, and air conditioning tests. The vehicle shall be tested to determine conformance to all sound level, heating, and air conditioning performance requirements specified in 3.2.3, 3.5.2.1, 3.5.10.1, and 3.5.11.1 respectively. Tests shall be as specified in 4.3.2.1 through 4.3.2.5. Test data, including a description of all applicable and pertinent test conditions, shall be recorded and furnished to the Government representatives. If the supplier has previously tested an identical model vehicle with the same components, a certification and the specified test data verifying conformance to this specification may be provided in lieu of the first vehicle tests. Certification and previous test results, if furnished, shall be forwarded to the engineering office of the procuring activity (USATACOM, AMSTA-UED).

4.3.2.1 Exterior sound level. The vehicle shall be tested in accordance with Part 325 of the regulations of the Department of Transportation to verify that the exterior sound level conforms to the Interstate Motor Carrier Noise Emission Standards of the Environmental Protection Agency.

4.3.2.2 Interior sound level (operator's location). The vehicle shall be tested to determine conformance to the interior sound level at the operator's location with driver in seated position, as specified in 3.2.3. Sound level shall be measured in accordance with procedures outlined in MIL-STD-1474 (Steady-State Noise, Personnel-Occupied Areas).

4.3.2.3 Interior sound level (passenger locations). Sound level shall be measured with all windows closed, at seated ear height, at various seat locations, and at designated vehicle speeds. A reading shall be taken at each location and speed specified (see 3.5.2.1). At 30 mph, the readings taken at the three positions shall be averaged to determine conformance to column A of table II. At the authorized maximum speed of the vehicle, the maximum reading at each position shall be taken to determine conformance to column B of table II. The sound measuring system shall consist of a condenser microphone, sound level meter, and an octave band analyzer which meet or exceed the instrumentation specified in MIL-STD-1474 (Steady-State Noise, Personnel-Occupied Areas). The sound measuring system shall be calibrated before the test and at the conclusion of the test.

* 4.3.2.4 Heating. To verify conformance to the heating performance requirements specified in 3.5.10.1, the supplier shall test the vehicle under actual or simulated conditions at least as stringent as specified herein or shall furnish engineering calculations certifying conformance to paragraph 3.5.10.1 at 0°F ambient temperature. The engineering calculations shall include:

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(a) Estimated engine heat output in Btu/hr and water temperature
(b) Total heater output in Btu/hr and water to ambient air temperature differential
(c) Body assembly heat loss in Btu/hr
(d) Estimated time required to raise the interior temperature from 0°F to 50°F with the bus pre-soaked in an ambient temperature of 0°F for 15 hours) when 170°F hot water is applied at the rate of three gallons per minute. (A 10-minute engine warmup time can be assumed.)
(e) No heat gain from solar radiation and passenger body heat
(f) Any significant heat losses.

Heating, defrosting and ventilating controls shall be operated to determine if they are functioning properly.

4.3.2.5 Air conditioning. To verify conformance to the air conditioning performance requirements specified in 3.5.11.1, the manufacturer shall test the vehicle under actual or simulated conditions at least as stringent as specified herein. Simulated conditions must be acceptable to the Government. Inside temperature may be calculated either with the specified number of seated passengers or through the use of simulated heat for sensible cooling. Solar radiation may be employed or simulated by calculating effective solar load in Btu, based on the square foot surface and the angle of incidence that will be exposed to the sun at the same time.

* 4.3.3 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and the distribution of the curb weight on the front and the rear axles. The imposed loading on the front and rear axles will be computed by the contractor and verified by the Government, using the curb weight and the payload, uniformly distributed over the vehicle load area, to provide the specified GVW. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle, and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

4.3.4 Road test. The first production vehicle shall be examined and road tested by the contractor in both bus and ambulance configurations, with payload, to assure that the vehicle will operate in accordance with the requirements herein.

4.3.4.1 Road test details. Unless otherwise specified (see 6.2), the first production vehicle shall be loaded to simulate the maximum payload and shall be subjected to a road test of not less than the mileage specified in table III. The test mileage shall be as indicated for the total quantity of vehicles of the same type (see 1.2) being procured under the contract at the time of contract award. One half of

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the road test shall be conducted with full payload allowance specified in 3.2.5 evenly distributed in the full passenger configuration; and one half of the road test shall be conducted with 300 pounds evenly distributed on each litter in the litter configuration. The test shall be conducted by the contractor under the direction of Government representatives. The test shall be conducted over a test course approved by the procuring activity and meeting the requirements of table IV. The vehicle shall be inspected periodically during and at the completion of the test by the contractor. The cost of the test and vehicle support, exclusive of personal expenses incurred by the participating Government representatives, shall be borne by the contractor.

4.3.4.1.1 Test course. The total test mileage shall be not less than the mileage specified in table III on roads and highways within the general geographic limits of the place of manufacture and adjoining states. The course shall be arranged to return the vehicle to the manufacturer's plant for inspection and chassis servicing at the completion of each 1,000 miles (and at the end of the test, if less than a 1,000 mile increment is required under table III). The turning radius requirement of 3.3.3 shall be verified at the completion of each 1,000 miles of road test required. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall preclude the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries.

TABLE III. Road test.

Vehicle contract quantity	Total test mileage
1- 5	250
6-10	1,000
11-20	2,500
21 or more	5,000

TABLE IV. Test course requirements.

Type of road	Percent of minimum miles
Dirt	2
Gravel	18
City streets	10
Unimproved black top	20
Improved pavement	50

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4.3.4.1.2 Test failure. The vehicle utilized for the road test shall successfully complete the test. Substitution of a new vehicle or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test vehicle shall be for deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels, frame, or driveshaft
(d) Vibration due to the type of body construction or mounting
(e) Failure of any drive line component
(f) Evidence of abnormal tire wear due to misalinement or unbalance
(g) Failure of any vehicular safety device such as brakes, steering assembly, windshield washers and wipers, or electrical circuits
(h) Evidence of structural weakness in any part of the vehicle, vehicle components, or accessories
(i) Loose mountings of parts or accessories due to workmanship or vehicular operation
(j) Excessive oil consumption (exceeding 1 quart per 1,000 miles)
(k) Failure or deformation of any part of the litter support system.

4.3.4.1.3 Mileage and maintenance records. The vehicle shall be driven not more than 1,000 miles more than the miles required under table III, including break-in mileage, prior to and during the test. All mileage accumulated and all service or maintenance records shall be furnished to the Government representatives. Distribution of records shall be identical to that for service manuals and shall be concurrent with the manual distribution.

4.3.4.1.4 Delivery of test vehicle. Upon successful completion of the vehicle test and final examination, the vehicle shall be cleaned, serviced and refurbished to the extent required for delivery of a new vehicle to the Government. Warranty provisions of the contract shall be exclusive of total test miles.

4.3.5 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

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4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. Each production vehicle, less payload, shall be examined and driven on approximately level road surfaces by the contractor to assure that the vehicle will operate in accordance with the contractual requirements. The contractor may authorize his subcontractor who will perform the final assembly work on the vehicle to make such examination and tests, but this shall not relieve the contractor of his obligation to deliver an acceptable vehicle or constitute a waiver of any warranty provision. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 has been accomplished.

4.4.1 Removal of instruments and controls. The requirements in 3.4.17 shall not be construed as requiring the removal of the instruments and controls as part of the final examination and test. However, the presentation of the complete vehicle with said instruments and controls in place shall be tantamount to certification that said instruments and controls are installed in a manner that facilitate their removal and servicing.

4.4.2 Rejection. Deficiencies of workmanship and nonconformance to any requirements herein shall be cause for rejection until corrective action has been taken.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for nontactical use by the Government in transporting both children and adults over roads and highways encountered by the military services in the continental United States and overseas areas, and for operation in all climates. Additionally, the buses are intended for conversion to ambulances for transportation of litter patients. The principal use of the vehicles will be stop-and-go, short haul, medium and low speed operation.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, and number of litters and number of passengers (see 1.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Concealed markings, if required (see 3.1.1.1).
(e) Gloss yellow exterior color, if required (see 3.1.1.1).
(f) Rustproofing, if required (see 3.1.1.2).
(g) Towing brakes, if required (see 3.1.1.9).
(h) Exterior sound level in accordance with California Vehicle Code, if required (see 3.2.3).
(i) Power plant heaters and fuel warmer, if required (see 3.4.1.7).
(j) If a battery heater is not required (see 3.4.1.7).
(k) Silicone rubber radiator and heater hoses, if required (see 3.4.1.8).
(l) Fuel fired engine preheater, if required (see 3.4.1.9).
(m) Passenger signal buzzer, if required (see 3.4.2.6.2).
(n) If red signal lamps are not required for type III vehicle (see 3.4.2.6.4).
(o) A master electrical switch, if required (see 3.4.2.7).
* (p) Special capacity protection for main fuel tank(s) serving air conditioner system, if required (see 3.4.3.2).
(q) Transmission lockup, if required for bus with fixed ratio type automatic transmission (see 3.4.5.1).
(r) Manual transmission, if required (see 3.4.5.2).
(s) Traction control, if required (see 3.4.9.1).
(t) Two-speed axle, if required (see 3.4.9.2).
(u) Disc type wheels, if required (see 3.4.10).
(v) Bias ply tires, if required (see 3.4.10.1).
(w) Carrier and spare wheel or rim assembly, if required (see 3.4.10.4).
(x) Carrier, spare wheel and tire assembly, if required (see 3.4.10.5).
(y) Air operated horn, if required (see 3.4.12).
(z) Tools, if required (see 3.4.16).
(aa) Right-hand drive, if required (see 3.4.17).
(ab) Rear step/seat, if required (see 3.5.3.4).
(ac) Rear loading ramp, if required (see 3.5.3.5).
(ad) Destination sign compartment, if required (see 3.5.4.4).
(ae) Sign roll with destinations, if required (see 3.5.4.4).
(af) Service door padlocking hasp and emergency door locking bolt, if required (see 3.5.5).
(ag) Air actuated doors, if required (see 3.5.5.1).

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(ah) Auxiliary heater, if required (see 3.5.10.1.1).
(ai) If air conditioning is not required (see 3.5.11).
(aj) Additional litter capacity for type III vehicles, if required (see 3.5.13.4).
(ak) Road test mileage, if different (see 4.3.4.1).

6.3 Performance prediction. Completed copies of SAE Truck Ability Predition Procedure form and computation for low speed requirement for each vehicle model furnished under contract should be submitted as specified in the contract. Unless other conditions are cited in this contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting vehicle ability (see 3.3.1.1) are established as follows for the corresponding SAE J688, Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2 - Altitude Factor. 1.00.
Table 3 - Rolling Factor. 1.841.
Table 4 - Area Factor. 0.197.
Table 5 - Velocity Factor. 333.
Table 6 - Altitude Factor. 1.00.
Table 7 - Chassis Friction Horsepower. Use applicable power unit GVW and the engine revolutions per minute (rpm) (to nearest 100 revolutions) which is required for a geared speed of 55 mph. For engine speed beyond the range of this table, the following formula shall be used: FHP = 1 + (0.0000002) (rpm) (GVW).
Table 8 - Grade Factor. 0.68.
Table 8A - Correction Factor. Not required.
Table 9 - Road factor. 0.0.

6.4 Subject term (key word) listing.
Bus
Bus, 30 or 44 passenger
Bus, air conditioning system
Bus, ambulance conversion
Bus, forward control model
Bus, heating system
Bus, high head room
Bus, litter transport
Bus, rear loading ramp
Bus, transit

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* 6.5 Changes in classification. In accordance with interservice agreement during coordination of MIL-B-45367 in May 1977, type I, 37- and 45-passenger, standard buses were deleted. For future procurement of forward control, 36- or 44-passenger, standard buses, use MIL-B-62261.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks (*) to indicate where changes (addition, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Preparing Activity:
Army - AT

Project No. 2310-0437

Review Activities:
Army - CR, ER, MD
Navy - MC
Air Force - 99

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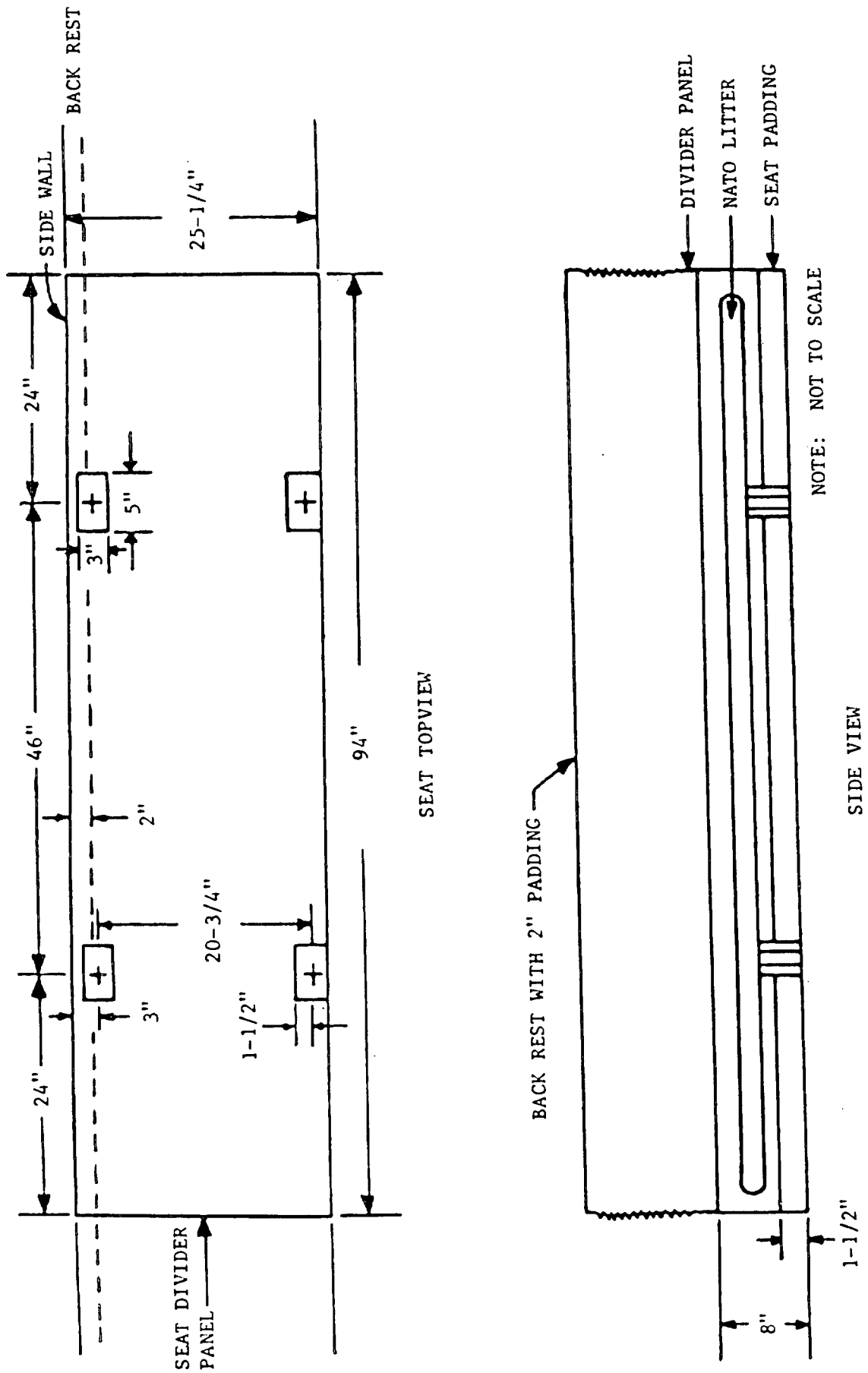


FIGURE 1. Schematic for litter cup placement - type IV bus.

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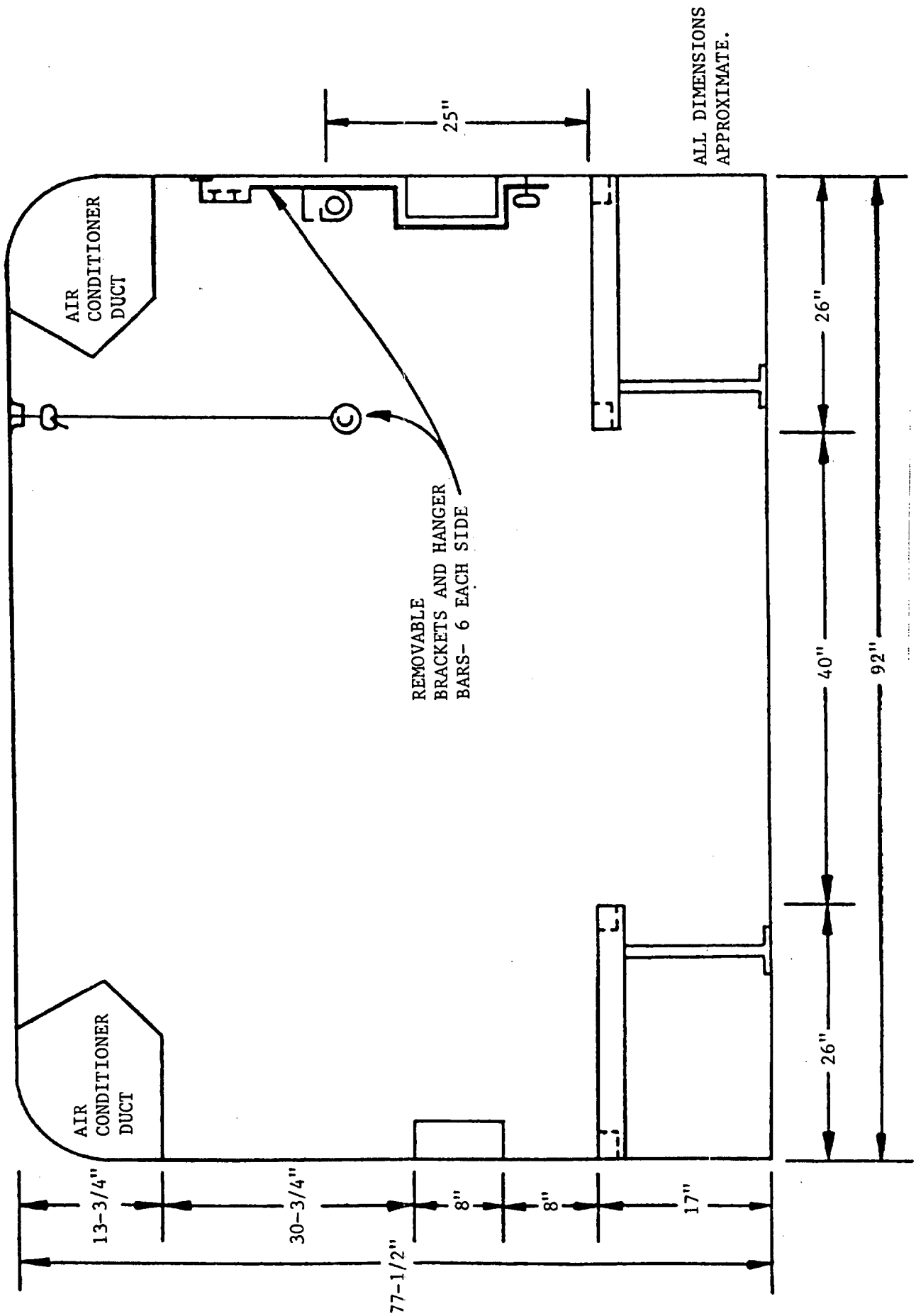


FIGURE 2. Litter support kit for type IV vehicle.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-45367 X	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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